

Point to Site VPN Work Instruction

PrimeCore Minerals | Secure Multi Office Network

PC-15 | Version 0.1 | 23 September 2026

Status: Proposed work instruction not executed

Owner: Cloud Administrator | Approver: Project lead | Approval date: Pending

Configure and validate remote user access to the proposed Azure private test workload. This instruction depends on PC-05 and does not assert an existing PrimeCore VPN service.

Prerequisites and parameter record

Have an approved Azure subscription and tenant, deployment rights, a supported route based VPN gateway and client, named test users, a verified MFA policy and an approved change window. Check current gateway capabilities and tenant requirements before deployment. Microsoft Entra authentication uses OpenVPN and Azure VPN Client. [M2]

Parameter	Proposed value or required entry
VNet and gateway	vnet-pc-lab 10.60.0.0/16; vgw-pc-lab
GatewaySubnet	10.60.255.0/27; no NSG
Client address pool	172.30.250.0/24; verify no local or project overlap
Test service	10.60.10.10 TCP 443; private workload
Authentication	Microsoft Entra ID with Microsoft-registered client
Tenant and issuer	Copy validated values from the actual tenant and current Microsoft guidance
Audience	Use current Microsoft-registered Azure VPN Client audience for the selected cloud
Authorised users and MFA	Record actual access assignment and effective authentication policy

Gateway configuration

1. Open the deployed gateway's point-to-site configuration. Record current settings before editing, including the address pool and allowed authentication methods.
2. Set the non-overlapping client pool, OpenVPN tunnel type and Microsoft Entra ID authentication. Enter the tenant, issuer and audience validated for this tenant and cloud. Do not copy identifiers from another organisation.
3. Complete the required tenant application consent and user-assignment configuration using an authorised identity administrator. Require assignment where supported and test that a non-approved user cannot connect.
4. Save the configuration and download a fresh VPN client profile after changes. Treat connection profiles as controlled configuration and distribute only through approved storage.

Client operation and verification

1. Install the supported Azure VPN Client on the authorised test endpoint. Import the downloaded Azure VPN profile and confirm the expected gateway and tenant.
2. Sign in with the named authorised account and complete the enforced authentication policy. Record the assigned address from 172.30.250.0/24 and inspect the route to 10.60.0.0/16.
3. Test the private HTTPS service by its configured name or address. If using a name, verify DNS resolution independently. Do not disable certificate validation to make an HTTPS test appear successful.
4. Test a prohibited port and an unapproved identity. Confirm that NSG rules, guest firewall and application permissions enforce the intended limits.
5. Disconnect and confirm the private workload is no longer reachable through that tunnel. For removal, revoke access and test new sign-in denial; assess active sessions separately and terminate them where supported.

Test ID	Expected result	Record
VPN-01 approved user	Connects, meets MFA policy and receives pool address	Actual identity and assigned IP
VPN-02 private HTTPS	Intended service responds with valid trust configuration	Service response and route
VPN-03 prohibited port	Denied despite connected VPN	Connection result and rule evidence
VPN-04 unapproved user	Cannot establish authorised session	Sign-in and access result
VPN-05 disconnect or revocation	Private access stops; new unauthorised connection denied	Time and separate session tests

Troubleshooting and rollback

If sign-in fails, check tenant, audience, issuer, consent, assignment and authentication policy. If connected but unreachable, check route overlap, assigned pool, effective workload NSG rules, guest firewall and service listener. If names fail but addresses work, investigate DNS. Collect diagnostic evidence without exposing tokens. Restore the previous gateway configuration if the change breaks approved access; regenerate and redistribute profiles if required.

This prototype reaches only the approved Azure workload. HQ access requires a separately configured and tested site-to-site path, return routes and firewall policy. Record client version, gateway SKU, test times and outcomes before marking the instruction as implemented.

References and linked artefacts

[S1] HQ_Network_Addressing_Artefact.md, project deliverables, evidence dated 13 September 2026.

[M1] Microsoft Learn. Create and manage a VPN gateway. <https://learn.microsoft.com/en-us/azure/vpn-gateway/tutorial-create-gateway-portal> (accessed 23 September 2026).

[M2] Microsoft Learn. Configure P2S VPN gateway for Microsoft Entra ID authentication. <https://learn.microsoft.com/en-us/azure/vpn-gateway/point-to-site-entra-gateway> (accessed 23 September 2026).